MIL-M-80127B
8 January 1980

SUPERSEDING
MIL-M-80127A
27 December 1973 and
MIL-M-80155A
4 December 1973

MILITARY SPECIFICATION

MILLING MACHINES, BED TYPE, RAM OR COLUMN MOUNTED VERTICAL SPINDLE, ELECTRONIC TRACER CONTROLLED, PROFILING AND DUPLICATING

1. SCOPE

1.1 Scope. This specification covers bed type milling machines with a vertical spindle mounted on a ram or a moving column. The machines are equipped with electronic controls for profiling and duplicating with equipment and accessories specified herein.

1.2 Classification. The specification covers machines of the following classes and sizes. The class and size required for a specific procurement shall be as specified (see 6.2.1).

Class A - 15 to 25 horsepower spindle drive

Size	Cross travel, inches (mm)	Longitudinal travel, inches (mm)
30-48	30 (762)	48 (1219.2)
40-96	40 (1016)	96 (2438.4)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, Memphis, Tennessee 38114, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 3417

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:40Z
Check the source to verify that this is the current version before use.

Class B - 30 or more horsepower spindle drive

Size	Cross travel, inches (mm)	Longitudinal travel, inches (mm)
32-60	32 (812.8)	60 (1524)
32-96	32 (812.8)	96 (2438.4)
32-120	32 (812.8)	120 (3048)
40-60	40 (1016)	60 (1524)
40-96	40 (1016)	96 (2438.4)
40-120	40 (1016)	120 (3048)
48-120	48 (1219.2)	120 (3048)
60-120	60 (1524)	120 (3048)
72-120	72 (1828.8)	120 (3048)

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

STANDARDS

FEDERAL

FED-STD-H28 - Screw-Thread Standards for Federal Services

MILITARY

MIL-STD-461 - Electromagnetic Interference Characteristics Requirement for Equipment
MIL-STD-1188 - Commercial Packaging of Supplies and Equipment

(Copies of specfications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

U. S. DEPARTMENT OF LABOR

OSHA 2206 - General Industry, OSHA Safety and Health Standards (29 CFR 1910)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:40Z
Check the source to verify that this is the current version before use.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

B5.1 - T-Slots, Their Bolts, Nuts and Tongues
B5.18 - Spindle Noses and Tool Shanks for Milling Machines
B11.8 - Construction, Care, and Use of Drilling, Milling, and Boring Machines, Safety Requirements for the
C113.1 - Electrical Standards for Metalworking Machine Tools (NFPA No. 79)
Z210.1 - Metric Practice (ASTM-E-380)

(Application for copies should be addressed to the American National Standards Institute, ATTN: Sales Dept., 1430 Broadway, New York, NY 10018.)

3. REQUIREMENTS

3.1 First article. When specified (see 6.2.1), the contractor shall furnish a sample unit for first article inspection and approval (see 6.2.1 and 6.3).

3.2 Design. The milling machine shall be new and one of the manufacturer's current models capable of accuractely duplicating from master models or templates, and performing conventional milling operations on a wide range of parts and dies in accordance with the requirements herein. The machine shall include all components, parts, and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair.

3.2.1 Measurement systems. Unless otherwise specified, either the U.S. Customary System of Units (US) or the International System of Units (SI) may be used in the design and construction of the machine. When only one system of measurements is acceptable, the particular system required shall be as specified (see 6.2.1). In this specification, all measurements, dimensions, sizes and capacities are given in the U. S. Customary System of Units (US). These measurements may be converted to the International System of Units (SI) through the use of the conversion factors and methods specified in ANSI Z210.1.

3.2.1.1 Measuring and indicating device calibrations. When specified (see 6.2.1), measuring and indicating devices such as feed screw dials, scales, depth stops, carriage stops, dial indicators, pressure gauges, temperature indicators, and all other similar devices shall be graduated in the specified system (US or SI) of measurements. Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:40Z
Check the source to verify that this is the current version before use.

3.2.1.2 Dual calibrations. When specified (see 6.2.1), measuring and indicating devices shall be graduated in both the US and the SI System of Measurements. When a dual US and SI system is furnished both feed dials shall have independent zero adjustments and both dials shall be calibrated in such a manner that the last dial graduation progresses into and is continuous with the frist dial graduation as the dial is rotated through the zero position.

3.2.2 Reclaimed materials. The machine shall contain reclaimed materials to the maximum extent possible without jeopardizing its intended use and performance. The reclaimed materials shall have been reprocessed, remanufactured or recycled in a manner which restores then to the same chemical composition and physical properties as the materials originally selected for use on the machine. Reclaimed materials shall include iron, steel, copper, brass, aluminum fiber products, plastics, and elastomers that have been collected from discarded solid, liquid, semi-solid, or gaseous waste such as garbage, refuse, or sludge.

3.2.3 Controls. All operating controls shall be located convenient to the operator at his normal work station.

3.2.4 Safety and health requirements. Covers, guards, or other safety devices shall be provided for all parts of the machine that present safety hazards. The safety devices shall not interfere with the operation of the machine. The safety devices shall prevent unintentional contact with the guarded part, and shall be removable to facilitate inspection, maintenance and repair of the parts. All machine parts, components, mechanisms, and assemblies furnished on the machine, whether or not specifically required herein, shall comply with all of the requirements of OSHA 2206 that are applicable to the machine itself. In addition, the machine shall comply with all requirements of ANSI B11.8 that are designated therein as the responsibility of the machine manufacturer. In the event of conflict between the requirements of the OSHA and ANSI Standards, the requirements of OSHA shall apply. Additional safety and health requirements shall be as specified (see 6.2.1 and 6.4).

3.2.5 Lubrication. Means shall be provided to insure adequate lubrication for all moving parts. Each recirculating system shall include a filter which is cleanable or replaceable. Each lubricant reservoir shall have means for determining fluid level. All oil holes, grease fittings, and filler caps shall be accessible.

3.2.6 Interchangeability. To provide for replacement of worn parts, all parts shall be manufactured to definite dimensions and tolerances.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:40Z
Check the source to verify that this is the current version before use.

3.3 Construction. The machine shall be constructed of parts which are new, without defects and free of repairs. The structure shall be capable of withstanding all forces encountered during operation of the machine to its maximum rating and capacity without permanent distortion.

3.3.1 Castings and forgings. All castings and forgings shall be free of scale and mismatching. No process such as welding, peening, plugging, or filling with solder or paste shall be used for reclaiming any defective part.

3.3.2 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. None of these operations shall be employed as a repair measure for any defective part.

3.3.3 Fastening devices. All screws, pins, bolts and other fasteners shall be installed in a manner that prevents change of tightness. Those subject to removal or adjustment shall not be swaged, peened, staked or otherwise permanently installed.

3.3.4 Surfaces. All surfaces shall be cleaned and free of sand, dirt, fins, sprues, flash, scale, flux and other harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.3.5 Painting. Unless otherwise specified (see 6.2.1), the machine shall be painted in accordance with the manufacturer's commercial practice.

3.3.6 Threads. All threaded parts used on the machine and its related attachments and accessories shall conform to FED-STD-H28 and the applicable "Detailed Standard" section referenced therein.

3.3.7 Dials. All dials that indicate stock removal or tool movement shall be graduated in increments not larger than 0.001 inch. Dial diameters shall be such that graduations may be read from the normal operating position. Dials shall be permanently and legibly engraved or etched on a non-glare background.

3.3.8 Plates. All words on instruction and indicating plates shall be in the English language. Characters shall be engraved, etched, embossed or stamped in boldface on a contrasting brackground.

3.3.9 Gears. All gears shall be machined from a material suitable for the intended purpose. All gears shall be heat treated by a process that will impart the hardness and toughness that will enable the gear train to transmit full rated torque of the drive motor without gear damage, failure or premature wear.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:40Z
Check the source to verify that this is the current version before use.

3.4 Components. The milling machine and control system furnished under this specification shall consist essentially of the following components.

3.4.1 Bed. The bed shall be a fixed height casting or weldment, ribbed for strength and rigidity. The bed shall contain the feed drive mechanism for the table and have the necessary access doors or covered opening to facilitate required adjustments and replacement of parts contained within. The length of the bed and bed ways shall be such that the worktable is full supported.

3.4.2 Table. The table shall have T-slots conforming to ANSI B5.1. It shall be accurately machined and fitted to the bedways. Channels and catch basins for chips and coolant shall be provided on the perimeter of the table with one or more openings for returning the coolant to a sump.

3.4.3 Column. The column shall serve as a base for the cross slide and have ways to provide for movement in the cross direction. The column shall be of sufficient strength and proportions to support the cross slide and sustain all forces imposed by full load machining operations. When column and table bed are constructed as separate units, the column shall be aligned, fitted, and firmly fastened to the foundation and table bed. The column shall have access doors as required for servicing the components located therein. All anchor bolts and leveling screws required by foundation plans shall be furnished with the machine.

3.4.4 Cross slide. The cross slide shall be constructed for strength and rigidity, with machined and finished slides to assure vibration free performance and shall have vertical ways square with the table.

3.4.5 Spindle head. The spindle carrier shall be a vertical sliding headstock, adequatedly propertioned to capably perform milling operations within the full range and capacity of the machine.

3.4.6 Spindle. Unless otherwise specified the machine shall have one spindle. When specified (see 6.2.1), the machine shall have multiple spindles. Each spindle shall be supported by two or more adjustable, precision roller or ball bearings and one other precision bearing for support of the spindle. The spindle shall have a precision ground Number 50 American National Standard Taper in accordance with ANSI B5.18. The spindle shall be provided with an arrangement for quick changing arbors and adapters without the necessity of a draw bar. When specified (see 6.2.1), a power draw bar shall be furnished. Controls shall be furnished to start, stop, jog, reverse and lock the spindle.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:40Z
Check the source to verify that this is the current version before use.

3.4.7 Way and way slides. The ways shall be properly spaced and be of suitable size to support movement of matching machine elements. The machine shall have simple means for adjustment to insure correct alignment of the mating components, and to compensate for wear of the mating surfaces.

3.4.8 Way wipers and covers. Wipers and covers or other means shall be provided to protect the way bearing surfaces from chips and other foreign matter.

3.4.9 Tracer system. The tracing mechanism shall be capable of controlling machine movement for precise duplication of master models or templates. A one to one relationship between the template or master model and the workpiece shall be maintained on finish cuts at all feed rates. The tracer mounting bracket shall be extended over the table, and attached to the spindle carrier or mounted on a separate master column. The bracket shall be adjustable in the longitudinal, cross and vertical planes to position the tracer stylus to the template or model. The tracer stylus shall respond to pressure light enough to permit the use of wood, plastic or other easily worked materials for master models. In addition to the above requirements, the machine shall have a tracer head meeting the requirements of 3.4.9.1.

3.4.9.1 Tracer head. The automatic 360° tracer head shall synchronize and control the movement in the following modes: automatic scan milling (Y, Z, or X, Z), automatic scan milling with Z-level reversal (Y, Z or X, Z), 360° automatic profile milling (X, Y), 360° profile milling with automatic Z axis incrementing (X, Y), 360° automatic profile milling with automatic depth control (X, Y, Z) with an angular capability of up to 25° from horizontal. The tracer shall be completely automatic once the tracer stylus has engaged the master and shall move in the selected direction. Means for hand guidance of the machine's longitudinal, transverse and depth movements shall be provided. Provisions shall be made for disengagement of the tracer mechanism in order that power feed control movements of table and cross slide may be made for conventional milling or set-up.

3.4.10 Coolant system. The coolant system shall be a flood or mist type as specified (see 6.2.1). The coolant system shall include a sump or reservoir, a power driven pump, and all necessary piping. The sump or reservoir shall have sufficient capacity to permit full flow of coolant. The system shall have means for draining and cleaning and shall include a strainer or baffle which is easily removable for cleaning. Means shall be provided to permit the operator to direct and control the amount of coolant.

3.4.11 Hydraulic system. The hydraulic system, when furnished, shall be complete, including all pumps, valves, piping cyclinders, and pressure

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:40Z
Check the source to verify that this is the current version before use.

controls. Overpressure protection shall be provided in the high pressure line to prevent damage to components. A filter system shall be provided to insure delivery of clean fluid. The hydraulic reservoir shall have means for determining fluid level.

3.4.12 Electrical system. Unless otherwise specified (see 6.2.1), the electrical system shall conform to all requirements in this paragraph. The system shall conform to the requirements of ANSI C113.1. Each machine shall draw all of its electrical power from a single 230/460 volt, 3 phase, 60 Hz circuit having a fused safety disconnect switch. The machine shall be initially wired for operation on 230 volts. An identified terminal for grounding the machine when installed shall be mounted in or near the disconnect switch. The terminal shall be suitable for connecting the size grounding conductor specified in Table 240-A of ANSI C113.1 for the disconnect fuse rating.

3.4.12.1 Motors. Motors shall have ball or roller bearings. Continuous duty horsepower ratings shall conform to the requirements of Table I. Unless otherwise specified (see 6.2.1), each motor enclosure shall meet the requirements for a dripproof enclosure.

3.5 Sizes and capacities. The sizes and capacities of the machine shall conform to the requirements of Table I. Optional sizes and capacities shall be furnished when specified (see 6.2.1).

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:40Z
Check the source to verify that this is the current version before use.

Table I Size and Capacities.

Characteristic	Class A		Class B								
	Size 30-48	Size 40-96	Size 32-60	Size 32-96	Size 32-120	Size 40-60	Size 40-96	Size 40-120	Size 48 x 120	Size 60 x 120	Size 72 x 120
Table:											
Travel, inches:											
Longitudinal	48	96	60	96	120	60	96	120	120	120	120
Cross	30	40	32	32	32	40	40	40	48	60	72
Vertical	16	16	24	24	24	24	24	24	24	30	30
Working surface, inches	36 x 52	40 x 100	28 x 83	28 x 119	28 x 143	40 x 83	40 x 119	40 x 143	40 x 120	60 x 120	60 x 120
Top of table to spindle nose, inches	14 to 28	18 to 32	6 to 30				6 to 30		8 to 32	25 to 55	25 to 55
T-Slots:											
Number	5	5	5			7			7	9	9
Size, inch	13/16	13/16	13/16						1-1/16	1-1/16	1-1/16
Distance between T-slots, inches	5	5	5						5	5	6
Spindle:											
Drive motor horsepower:											
Standard	15	20	30						40		
Speeds, Number of	16	16	24						infinite		
Speed range, RPM:											
Standard	50 to 3000	50 to 3000	20 to 2000						18 to 1800		
Optional			16 to 1600								
			24 to 2400								
Feed rates:											
Number of	infinite	infinite	infinite						infinite		
Range, tracer controlled, IPM:											
Longitudinal	1 to 40	1 to 40	1 to 50						1 to 50		
Cross	1 to 40	1 to 40	1 to 50						1 to 50		
Vertical	1 to 40	1 to 40	1 to 50						1 to 50		
Rapid traverse, Longitudinal and cross, IPM	100	120	175						100		

NOTE: Requirements are not less than those shown. Where a range is shown, the required performance is from the smaller figure, or less, to larger figure, or more.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:40Z
Check the source to verify that this is the current version before use.

MIL-M-80127B

3.6 Performance. Each machine shall be capable of satisfactory performance when tested in accordance with 4.5.3 through 4.5.4.

3.7 Alignment accuracies. Each machine shall meet the alignment accuracies of Table II.

TABLE II. Alignment accuracies.

Description	Allowable deviation not more than
Spindle end play:	
Taper bearings @ 1000 lbs. applied force	0.001 inch
Ball bearings @ 650 to 850 lbs. applied force	0.0002 inch
Spindle taper runout:	
2 inches from spindle nose	0.0005 inch TIR
12 inches from spindle nose	0.001 inch TIR
Vertical movement of spindle head square with table top	0.0015 inch per foot
Spindle square with table top:	
Single spindle	0.002 inch per foot
Multiple spindles	0.002 inch per foot
Transverse movement of spindle head parallel to table top	0.002 inch per foot
T-slots parallel to full length table travel	0.0008 inch per foot
Rise and fall of table in full length of travel	
48 inches table travel	0.002 inch
60 inches table travel	0.002 inch
96 inches table travel	0.002 inch
120 inches table travel	0.003 inch
Cross slide movement square with T-slots	0.0008 inch per foot

3.8 Standard equipment. Unless otherwise specified (see 6.2.1), the following equipment shall be furnished with each machine.

An electrical load indicator to indicate the electrical load on each spindle drive motor.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:40Z
Check the source to verify that this is the current version before use.

3.9 Optional equipment. Optional equipment shall be furnished as specified and fully described (see 6.2.1).

3.10 Additional requirements. Performance features required in addition to, or in lieu of, the requirements herein shall be as specified (see 6.2.1).

3.11 Electromagnetic interference control. When specified (see 6.2.1), equipment procured under this specification shall comply with the requiremetns of MIL-STD-461.

3.12 Fungus control. When required (see 6.2.1), fungus proofing shall be as specified.

3.13 Lubricating chart or plate. Unless otherwise specified (see 6.2.1), a lubrication chart or plate shall be permanently and securely attached to each machine. If a chart is furnished, it shall be placed in a transparent plastic folder, or permanently sealed between clear plastic sheets, with suitable means for mounting. The chart or plate shall contain the following information:

Points of lubricant application
Service interval
Type of lubricant
Viscosity

3.14 Nameplate. Unless otherwise specified (see 6.2.1), a corrosion resistant metal nameplate shall be securely attached to each machine. The nameplate shall contain the information listed below. If the machine is a special model, the model designation shall include the model of the basic standard machine and a suffix identified in the manufacturer's permanent records. The captions listed may be shortened or abbreviated, provided the entry for each caption is clear as to its identity.

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input (volts, total amps, phase, frequency)
Contract Number or Order Number
National Stock Number or Plant Equipment Code
Date of manufacture
U S

3.15 Technical data. Technical data shall be furnished as specified (see 6.2.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:40Z
Check the source to verify that this is the current version before use.

3.16 Workmanship. Workmanship of the machine and accessories shall be of a quality equal to that of the manufacturer's commercial equipment of the type specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of theinspection requirments specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification when such action is deemed necessary to assure supplies and services conform to perscribed requirements.

4.2 First article inspection. When first article approval is required under 3.1, first article inspection shall be performed. Unless otherwise specified (see 6.2.1), first article inspection shall comprise the examination in 4.4 and all tests in 4.5. Failure of the item to pass any examination or test shall be cause for disapproval of the first article.

4.3 Quality conformance inspection. Each item shall be subjected to quality conformance inspection prior to being offered for acceptance. Unless otherwise specified (see 6.2.1), quality conformance inspection shall consist of the examination in 4.4, the tests in 4.5.1 through 4.5.4 and the inspection in 4.6. Failure of the item to pass any examination, test, or inspection shall be cause for rejection.

4.4 Examination. The machine and equipment shall be examined to determine compliance with the requirements in 3.2 through 3.5 and 3.8 through 3.16.

4.5 Tests.

4.5.1 Operational test. The machine shall be operated at no load for not less than 3 hours. Proper operation of all controls, motors, adjusting mechanisms, and accessories shall be verified during the trial period.

4.5.2 Alignment accuracy tests. Each machine shall be tested to determine that it complies with the alignment accuracy requirements of Table II.

4.5.3 Horsepower, milling accuracy and tool chatter tests. Each machine shall be tested by machining a workpiece of SAE 4140 steel. The workpiece shall be not less than 12 inches long by 6 inches wide by 4 inches thick. One of the two surfaces having the largest area shall be ground flat and mounted in contact with the table. The workpiece may

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:40Z
Check the source to verify that this is the current version before use.

be drilled to facilitate clamping. The machine shall make roughing cuts to mill metal off the horizontal and one vertical surface of the workpiece. The roughing cut on the horizontal surface shall be made at sufficient depth and feed rate to load the spindle drive motor to not less than maximum continuous duty rated electrical input. Type of cutting tool, depth of cut, spindle speed and feed rate shall be determined by the manufacturer, provided the required loading is achieved. There shall be no evidence of chatter, stalling or permanent distortion during or after the test. Finishing cuts shall be made on the horizontal and vertical faces. The finished surfaces shall meet the following accuracies:

4.5.3.1 Squareness. The two milled sides shall be tested for squareness. The sides shall be square with each other within 0.001 inch per four inches.

4.5.3.2 Flatness. The two milled sides shall be tested for flatness. The sides shall be flat within 0.001 inch concave only.

4.5.3.3 Parallelism. The finished horizontal surface shall be tested for parallelism. The surface shall be parallel to the worktable within 0.001 inch total.

4.5.4 Tracer test. Each machine shall be tested in accordance with the manufacturer's standard tracer test provided that the test proves the capability of the machine to meet all of the requirements for which it was designed.

4.5.5 Electromagnetic interference control tests. Equipment requiring electromagnetic interference control testing shall be tested for compliance with 3.11.

4.6 Packaging inspection. Packaging shall be inspected to determine compliance with the requirements of Section 5.

5. PACKAGING

5.1 Preservation, packing and marking. Unless otherwise specified (see 6.2.1), preservation, packing and marking shall conform to the requirements of MIL-STD-1188.

6. NOTES

6.1 Intended use. The milling machines covered by this specification are to be used for conventional milling operations and manual or automatic profiling and duplicating of a wide range of parts.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:40Z
Check the source to verify that this is the current version before use.

6.2 Ordering data.

6.2.1 Procurement requirements. Purchasers should specify their requirements in procurement documents, including whether each choice is required or not required, by entering an appropriate statement identified to each of the following:

a. Title, number and date of this specification.
b. Class and size required (see 1.2).
c. When a first article is required for inspection and approval (see 3.1, 4.2 and 6.3).
d. If machine is required to be configured in a specific measurement system (US or SI), state required system (see 3.2.1).
e. *If measuring and indicating devices are required to be* graduated in a specific measurement system (US or SI), state required system (see 3.2.1.1).
f. Dual calibrations (US and SI), if required (see 3.2.1.2).
g. Specify additional safety and health requirements (see 3.2.4).
h. Painting, if different (see 3.3.5).
i. Multiple spindle machine, if required and number of spindles required (see 3.4.6).
j. Power draw-bar, if required (see 3.4.6).
k. Specify flood or mist type coolant system (see 3.4.10).
l. Electrical system, if different (see 3.4.12).
m. Type motor enclosure, if different (see 3.4.12.1).
n. Optional sizes and capacities, if required (see 3.5 and Table I).
o. Standard equipment, if different (see 3.8).
p. Optional equipment, if required; fully describe required optional equipment (see 3.9).
q. Specify additional features required (see 3.10).
r. Electromagnetic interference control, if required (see 3.11).
s. Type fungus proofing, if required (see 3.12).
t. Lubrication chart or plate, if different (see 3.13).
u. Nameplate, if different (see 3.14).
v. First article inspection, if different (see 4.2).
w. Quality conformance inspection, if different (see 4.3)
x. Preservation, packing and marking, if different (see 5.1).

6.2.2 Contract data requirements. Required technical data, such as operators manuals, parts lists, wiring diagrams, and other instructions for operation and maintenance as identified on a numbered DD Form 1664 should be specified on a DD Form 1423 incorporated into the contract.

6.3 First article. When a first article is required, it shall be tested and approved under the appropriate provisions of 7-104.55 of the

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:40Z
Check the source to verify that this is the current version before use.

Armed Services Procurement Regulation. The first article should be a preproduction sample, or a standard production item from the contractor's current inventory. The contracting officer should include specific instructions in all procurement instruments regarding arrangements for examinations, test and approval of the first article.

6.4 Safety and health requirements. Paragraph 3.2.4 requires compliance only with those OSHA requirements that concern the machine itself. It does not require compliance with those OSHA requirements that concern "the machine in its operating environment" such as noise levels, radiation levels, electromagnetic emissions, noxious vapors, air contaminants, heat, etc, Since OSHA limits the total hazard level of these hazards in the environment (and does not limit the hazard level of individual machines in the environment) the requiring agency is advised to analyze the existing hazards levels in the propesed operating environment, and specify additional machine requirements that will integrate the new machine into its future operating environment. If specific point-of-operation guarding is required, the requiring agency should specify the exact configuration of the guard required as in most cases the guard configuration is dependent on the size and configuration of the workpieces that will be machined. The above, and any other additional safety and health requirements should be specified in detail under 6.2.1(g).

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - AL
Air Force - 99

Review activities:
Army - AL
DLA - GS

User activities:
Navy - MC

Preparing activity:
DLA - IP

Project Number:
3417-0150

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:40Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS: This form is provided to solicit beneficial comments which may improve this document and enhance its use. DoD contractors, government activities, manufacturers, vendors, or other prospective users of the document are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity. A response will be provided to the submitter, when name and address is provided, within 30 days indicating that the 1426 was received and when any appropriate action on it will be completed.
NOTE: This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

DOCUMENT IDENTIFIER *(Number)* AND TITLE
MIL-M-80127B

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

SUBMITTED BY *(Printed or typed name and address — Optional)*	TELEPHONE NO.
	DATE

DD FORM 1426 1 OCT 76

EDITION OF 1 JAN 72 MAY BE USED UNTIL EXHAUSTED

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:40Z
Check the source to verify that this is the current version before use.